BOUNDED HOMOTOPY CATEGORIES QUOTIENTED BY ACYCLICS GIVE THE BOUNDED DERIVED CATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4063-4109, following the statement at 4021-4061.

Editorial note. Part (2) is 'dual to the proof of (1)' (4101) and part (3) reuses 'the argument given in the previous paragraph' with bounded-above complexes (4106-4108): mirror-level and direct-adaption omissions, recorded.

Allowed standard prerequisites (editorial).

- Acyclics as kernel of a homological functor. lemma-acyclic-general, cited at proof line 4065; complete proof at source lines 2465-2489, queued for separate carding. Supplies that Ac^+ etc. are strictly full saturated triangulated subcategories with Qis as the corresponding multiplicative system, via H^0.
- Boundedness adjustment of complexes. lemma-complex-cohomology-bounded (carded L26110, S9-der-truncation-bounded-models), cited at proof lines 4070, 4081, 4092. Supplies essential surjectivity and the two quasi-isomorphism adjustments (L' -> L" bounded below) used in fullness and faithfulness.

Source reference. Original lemma 05RM: sources/stacks-derived/pinned-original-source.tex, source lines 2448–2448. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05RV: sources/stacks-derived/pinned-original-source.tex, source lines 3977–3977. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05RL: sources/stacks-derived/pinned-original-source.tex, source lines 2395–2395. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. The subcategories $Ac^+(\mathcal{A})$, $Ac^-(\mathcal{A})$, resp. $Ac^b(\mathcal{A})$ are strictly full saturated triangulated subcategories of $K^+(\mathcal{A})$, $K^-(\mathcal{A})$, resp. $K^b(\mathcal{A})$. The corresponding saturated multiplicative systems (see Lemma 05RL) are the sets $Qis^+(\mathcal{A})$, $Qis^-(\mathcal{A})$, resp. $Qis^b(\mathcal{A})$.*

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *The kernel of the functor* $K^+(\mathcal{A}) \to D^+(\mathcal{A})$ *is* $Ac^+(\mathcal{A})$ *and this induces an equivalence of triangulated categories*
$$K^+(\mathcal{A})/Ac^+(\mathcal{A}) = Qis^+(\mathcal{A})^{-1}K^+(\mathcal{A}) \longrightarrow D^+(\mathcal{A})$$
(2) *The kernel of the functor* $K^-(\mathcal{A}) \to D^-(\mathcal{A})$ *is* $Ac^-(\mathcal{A})$ *and this induces an equivalence of triangulated categories*
$$K^-(\mathcal{A})/Ac^-(\mathcal{A}) = Qis^-(\mathcal{A})^{-1}K^-(\mathcal{A}) \longrightarrow D^-(\mathcal{A})$$
(3) *The kernel of the functor* $K^b(\mathcal{A}) \to D^b(\mathcal{A})$ *is* $Ac^b(\mathcal{A})$ *and this induces an equivalence of triangulated categories*
$$K^b(\mathcal{A})/Ac^b(\mathcal{A}) = Qis^b(\mathcal{A})^{-1}K^b(\mathcal{A}) \longrightarrow D^b(\mathcal{A})$$

Proof. The initial statements follow from Lemma 05RM by considering the restriction of the homological functor H^0. The statement on kernels in (1), (2), (3) is a consequence of the definitions in each case. Each of the functors is essentially surjective by Lemma 05RV. To finish the proof we have to show the functors are fully faithful. We first do this for the bounded below version.

Suppose that $K^\bullet, L^\bullet$ are bounded below complexes. A morphism between these in $D(\mathcal{A})$ is of the form $s^{-1}f$ for a pair $f : K^\bullet \to (L')^\bullet$, $s : L^\bullet \to (L')^\bullet$ where s is a quasi-isomorphism. This implies that $(L')^\bullet$ has cohomology bounded below. Hence by Lemma 05RV we can choose a quasi-isomorphism $s' : (L')^\bullet \to (L'')^\bullet$ with $(L'')^\bullet$ bounded below. Then the pair $(s' \circ f, s' \circ s)$ defines a morphism in $\mathrm{Qis}^+(\mathcal{A})^{-1}K^+(\mathcal{A})$. Hence the functor is "full". Finally, suppose that the pair $f : K^\bullet \to (L')^\bullet$, $s : L^\bullet \to (L')^\bullet$ defines a morphism in $\mathrm{Qis}^+(\mathcal{A})^{-1}K^+(\mathcal{A})$ which is zero in $D(\mathcal{A})$. This means that there exists a quasi-isomorphism $s' : (L')^\bullet \to (L'')^\bullet$ such that $s' \circ f = 0$. Using Lemma 05RV once more we obtain a quasi-isomorphism $s'' : (L'')^\bullet \to (L''')^\bullet$ with $(L''')^\bullet$ bounded below. Thus we see that $s'' \circ s' \circ f = 0$ which implies that $s^{-1}f$ is zero in $\mathrm{Qis}^+(\mathcal{A})^{-1}K^+(\mathcal{A})$. This finishes the proof that the functor in (1) is an equivalence.

The proof of (2) is dual to the proof of (1). To prove (3) we may use the result of (2). Hence it suffices to prove that the functor $\mathrm{Qis}^b(\mathcal{A})^{-1}K^b(\mathcal{A}) \to \mathrm{Qis}^-(\mathcal{A})^{-1}K^-(\mathcal{A})$ is fully faithful. The argument given in the previous paragraph applies directly to show this where we consistently work with complexes which are already bounded above. □